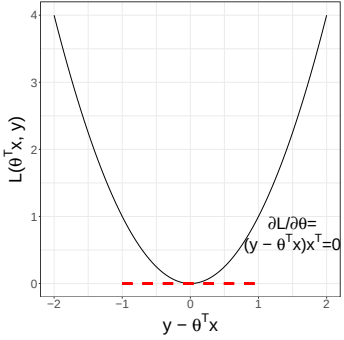
# Introduction to Machine Learning

# Supervised Regression In a Nutshell

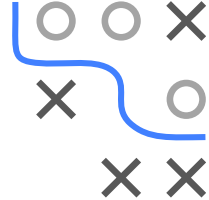



**Learning goals**

- Understand basic concept of regressors
- Understand difference between L1 and L2 Loss
- Know basic idea of OLS estimator

# LINEAR REGRESSION TASKS

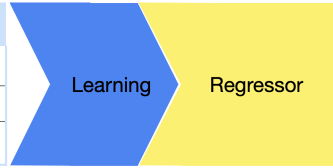
- Learn linear combination of features for predicting the target variable
- Find best parameters of the model by training w.r.t. a loss function
  $CreditBalance = \theta_0 + \theta_1 Rating + \theta_2 Income + \theta_3 CreditLimit$



Training

Input: Labeled data

| Rating | Income | Credit Limit | Credit Card Balance |
|---|---|---|---|
| 283 | 14.891 | 3606 | 333 |
| 483 | 106.025 | 6645 | 903 |
| 514 | 104.593 | 7075 | 580 |

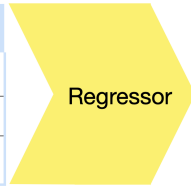



Output

Prediction

Input: Unlabeled data

| Rating | Income | Credit Limit | Credit Card Balance |
|---|---|---|---|
| 107 | 32.318 | 4351 | ? |
| 471 | 88.180 | 5042 | ? |
| 512 | 121.218 | 8101 | ? |

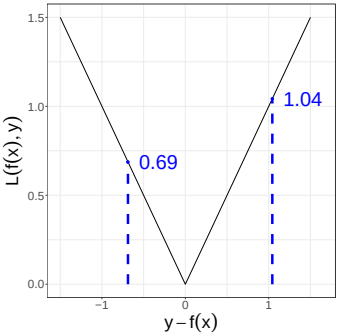


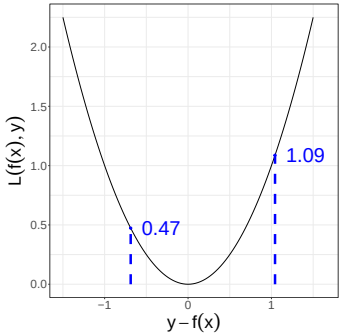

Prediction

| Credit Card Balance |
|---|
| 482 |
| 720 |
| 987 |

# LINEAR MODELS: L1 VS L2 LOSS

Loss can be characterized as a function of residuals $r = y - f(\mathbf{x})$



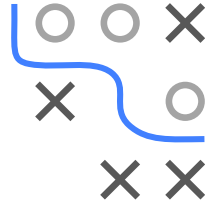

- **L1** penalizes the **absolute** value of residuals
- $L(r) = |r|$
- Robust to outliers

- **L2** penalizes the **quadratic** value of residuals
- $L(r) = r^2$
- Easier to optimize

# LINEAR MODELS: L1 VS L2 LOSS

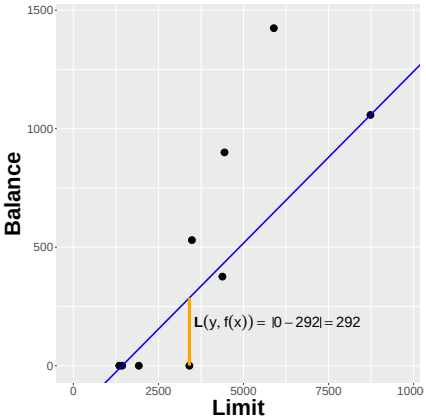
- **L1** Loss is not differentiable in $r = 0$
- Optimal parameters are computed numerically

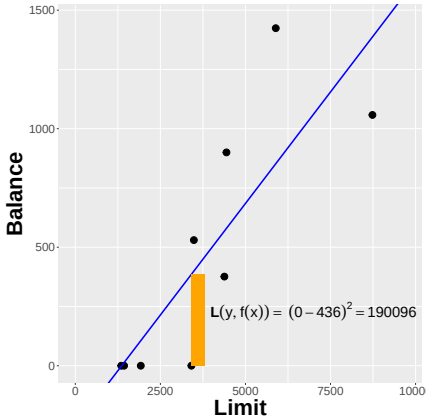
- **L2** is a smooth function hence it is differentiable everywhere
- Optimal parameters can be computed analytically or numerically



# LINEAR MODELS: L1 VS L2 LOSS

- The parameter values of the best model depend on the loss type



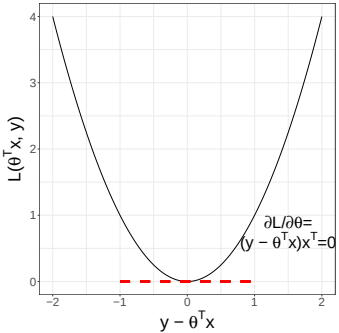




- $\hat{\theta}_{L_1} = 0.14 \rightarrow$ if the Credit Limit increases by 1\$ the Credit Balance increases by 14 Cents
- $\hat{\theta}_{L_2} = 0.19 \rightarrow$ if the Credit Limit increases by 1\$ the Credit Balance increases by 19 Cents

# OLS ESTIMATOR

**Ordinary-Least-Squares (OLS)** estimator:

- Analytical solution for linear models with L2 loss
- Best parameters can be computed via derivation of the empirical risk
- Solution: $\hat{\boldsymbol{\theta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$



# OLS ESTIMATOR

Components of **OLS** estimator:

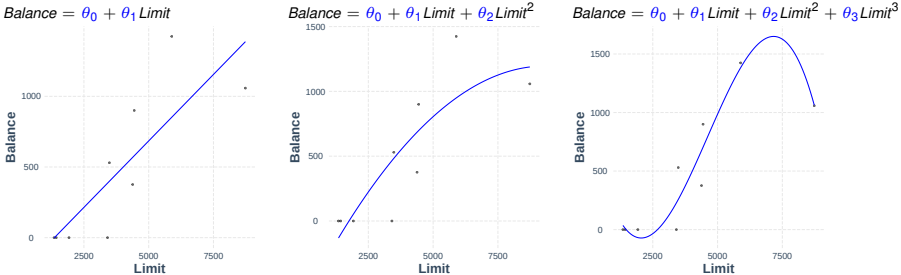
- **X**: Features + extra column for intercept
- **y**: Label vector

$X$

| Intercept | Rating | Income | Credit Limit |
|---|---|---|---|
| 1 | 283 | 14.891 | 3606 |
| 1 | 483 | 106.025 | 6645 |
| 1 | 514 | 104.593 | 7075 |

$y$

| Credit Card Balance |
|---|
| 333 |
| 903 |
| 580 |

# POLYNOMIAL REGRESSION

- Adding polynomial terms to the linear combination leads to more flexible regression functions
- Too high degrees can lead to overfitting